

Note on p -adic Expansions in $\mathbb{Q}$

Here we tie up some loose ends concerning how p -adic expansions of (ordinary) integers and rationals look inside of $\mathbb{Q}_p$. In what follows, we let $\mathbb{N}$ denote the positive integers and let $p \in \mathbb{N}$ denote an arbitrary prime. Let us also call a sequence $(a_n)_{n \in \mathbb{N}}$ *periodic* if and only if there exists an $s \in \mathbb{N}$ with $a_n = a_{n+s}$ for all $n \in \mathbb{N}$. The least (positive) such s is called *the period of* $(a_n)_{n \in \mathbb{N}}$. We call $(a_n)_{n \in \mathbb{N}}$ *eventually periodic* (or *ultimately periodic*) if and only if there exists a $k \in \mathbb{N}$ with $(a_{n+k})_{n \in \mathbb{N}}$ periodic. If k is the minimal nonnegative integer with this property then we call $(a_1, \dots, a_{k-1})$ the *pre-period*, and $(a_k, a_{k+1}, \dots)$ the *periodic part*, of $(a_n)_{n \in \mathbb{N}}$. Finally, abusing notation slightly, for any p -adic rational $x = \sum_{n=j}^{\infty} a_n p^n$, we call the sequence $(a_j, a_{j+1}, \dots)$ the *digits of the p -adic expansion of x* . We also say that x has a *finite p -adic expansion* if and only if $a_n = 0$ for sufficiently large n .

From basic arithmetic, we know that positive integers always admit finite base- b expansions for any $b \in \mathbb{N}$. So it is easily checked that $x \in \mathbb{Q}_p$ has a finite p -adic expansion if and only if $x = \frac{N}{p^k}$ for some $k, N \in \mathbb{N} \cup \{0\}$. So let us now observe that negative integers have rather distinctive (non-finite) p -adic expansions. (Thanks to Nathan Melhop for pointing this fact out in class!)

Lemma 1 *The periodic part of the digits of the p -adic expansion of any negative integer x is $(p-1, p-1, \dots)$.*

Proof: First observe that $-1 = \frac{p-1}{1-p}$, and the latter fraction admits a convergent power series expansion in $\mathbb{Z}_p$: $(p-1) \sum_{n=0}^{\infty} p^n$. So the lemma certainly holds for $x = -1$.

So assume $x \in \mathbb{Z}$ satisfies $x \leq -2$. Clearly, the *positive* integer $-x-1$ admits a finite p -adic expansion $a_0 + \dots + a_{k-1} p^{k-1}$, with $k = 1 + \lceil \log_p -x \rceil$. Then $x = x+1-1 = x+1 + \sum_{n=0}^{\infty} (p-1)p^n$, and the latter number is simply

$$\begin{aligned} & -(-x-1) + \left(\sum_{n=0}^{k-1} (p-1)p^n \right) + \sum_{n=k}^{\infty} (p-1)p^n \\ & \left(\sum_{n=0}^{k-1} a_n p^n \right) + \left(\sum_{n=0}^{k-1} (p-1)p^n \right) + \sum_{n=k}^{\infty} (p-1)p^n \end{aligned}$$

Since $|x+1| \leq \sum_{n=0}^{k-1} (p-1)p^n$, the sum of the first two terms is in fact a nonnegative integer, and thus has a finite p -adic expansion. So the periodic part of x is indeed as stated. ■

We next prove a more general observation from class.

Lemma 2 *A p -adic rational number x lies in $\mathbb{Q}$ (i.e., is a ratio of ordinary integers) if and only if the p -adic expansion of x is eventually periodic.*

Proof: ($\implies$): The statement is clear for $x \in \mathbb{Z}$, so let us assume $x \in \mathbb{Q} \setminus \mathbb{Z}$. Since multiplying x by integer powers of p only shifts p -adic digits, we may also assume $\text{ord}_p x = 0$. So we may assume $x = \frac{M}{N}$ with $M, N \in \mathbb{Z}$ and $\gcd(M, p) = \gcd(N, p) = 1$. Note then that the congruence $p^k \equiv 1 \pmod{N}$ has a solution $k \in \{1, \dots, N-1\}$, thanks to Euler's Theorem. In particular, we can write $p^k \equiv 1 - vN$ for some $v \in \mathbb{Z}$. So then, $x = \frac{M}{N} = \frac{vM}{vN} = \frac{vM}{1-p^k}$, and we will soon see that k is a multiple of the period of expansion we seek.

To continue, set $R := vM$ and, if $R \neq -1$, let j be the least nonnegative integer such that $p^j > R$. (If $R = -1$ then set $j = 0$.) If R is a positive multiple of $p^k - 1$ then set $c = p^k - 1$. Otherwise, let $c \in \{0, \dots, p^k - 2\}$ be the unique solution of $c \equiv \frac{R}{p^j} \pmod{p^k - 1}$. Setting $y := \frac{R - cp^j}{1 - p^k}$ we then see that y is an integer in $\{0, \dots, p^j - 1\}$, with base- p expansion $y_0 + y_1 p + \dots + y_{j-1} p^{j-1}$. We also see that $R = (1 - p^k)y + p^j c$, so that $x = y + \frac{p^j c}{1 - p^k}$. In particular, we obtain that the sequence of digits of the

p -adic expansion of x is exactly

$$(y_0, \dots, y_{j-1}, c_0, \dots, c_{k-1}, c_0, \dots, c_{k-1}, \dots),$$

where $c_0 + c_1p + \dots + c_{k-1}p^{k-1}$ is the base- p expansion of c . ■

($\Leftarrow$): By assumption, we can write x in the form $\left(\sum_{n=j}^{k-1} a_n p^n\right) + \sum_{n=k}^{\infty} a_n p^n$, where $j \leq k-1$, and $(a_k, a_{k+1}, \dots)$ is periodic with period s . In particular, letting $x_0 := \sum_{n=j}^{k-1} a_n p^n$, we obtain that x is the sum of the (ordinary) rational number, x_0 , and the number

$$\begin{aligned} z &:= \sum_{i=0}^{\infty} (a_{k+si} p^{k+si} + \dots + a_{k+s-1+si} p^{k+s-1+si}) \\ &= (a_k p^k + \dots + a_{k+s-1} p^{k+s-1}) \sum_{i=0}^{\infty} p^{si} \\ &= (a_k p^k + \dots + a_{k+s-1} p^{k+s-1}) \sum_{i=0}^{\infty} p^{si} \\ &= \frac{a_k p^k + \dots + a_{k+s-1} p^{k+s-1}}{1 - p^s}, \end{aligned}$$

which is clearly rational. So we are done. ■